CHAPTER 11

Network Address Translation

Private addresses · NAPT / PAT · translation table · inbound traffic · what NAT changes

QUICK TRICK

Outgoing: NAT replaces **(private IP, private port)** by **(public IP, new port)** and stores the pair in a table. Incoming reply: look the port up and put the private pair back. One public IP can carry up to about 2^{16} = **65,536** connections.

11.1 Why NAT?

● **LEVEL 1 · BEGINNER**

The problem. There are not enough public IPv4 addresses for every device. **The idea.** Give each home or office **private** addresses (10.0.0.0/8, 172.16.0.0/12, 192.168.0.0/16) and let one router share a **single public address**.

Analogy. A company reception desk. Staff have internal extension numbers. All outside calls show the one company number, and the receptionist remembers who is talking to whom.

11.2 How NAT works

● **LEVEL 2 · EXAM**

```
   Inside (private)                NAT router                    Outside (Internet)
Host 10.0.0.5:4000  --->  src 10.0.0.5:4000   ==>   src 203.0.113.1:5001  ---> Server 198.51.100.9:80
                          table: (10.0.0.5, 4000) <-> (203.0.113.1, 5001)
Host 10.0.0.5:4000  <---  dst 10.0.0.5:4000   <==   dst 203.0.113.1:5001  <--- reply from server
```

Direction	What NAT changes	What NAT keeps
Inside → outside	Source IP and source port; IP and TCP/UDP checksums	Destination IP and port
Outside → inside	Destination IP and destination port; checksums	Source IP and port

- **NAPT / PAT** (the usual kind): many private hosts share one public IP and are told apart by the **external port**.
- **Static NAT:** one private ↔ one public address, fixed. **Dynamic NAT:** private hosts take addresses from a pool of public IPs.
- Two inside hosts may pick the **same** source port. NAT must give them **different** external ports.
- Table entries are created by **outgoing** traffic and time out when idle.

11.3 Worked example

● **LEVEL 2 · EXAM**

NAT table of a router whose public IP is 203.0.113.1:

Inside (private IP : port)	Outside (public IP : port)
10.0.0.2 : 3000	203.0.113.1 : 6000
10.0.0.3 : 3000	203.0.113.1 : 6001

A packet arrives from 198.51.100.7:80 addressed to **203.0.113.1:6001**. NAT finds row 2 and rewrites the destination to **10.0.0.3:3000**; the source is unchanged. A packet to 203.0.113.1:7777 has no entry, so it is dropped.

11.4 Limits and side effects

● **LEVEL 2 · EXAM**

- **Inbound connections** from outside fail unless an entry exists (port forwarding / static mapping).

- **Breaks the end-to-end idea**: the router edits transport headers; protocols that carry IPs or ports inside the data (e.g. active FTP) need special help.
- Connection limit: a port field is 16 bits, so ≈ **65,536** simultaneous translations per public IP (fewer if low ports are reserved).
- Does **not** by itself provide security, though outsiders cannot start connections to hidden hosts.

11.5 Common mistakes

COMMON MISTAKE

- Saying NAT changes the **destination** on the way out (it changes the **source**).
- Forgetting that the reply comes back to the **public** address and port, and is translated back.
- Giving two inside hosts the same external port.
- Forgetting that the checksums must be recomputed.

QUICK TRICK

Memory line: "Out: change the source. In: change the destination. The port is the key."

GATE QUICK CHECK — Chapter 11

```
Out : (src_ip, src_port) -> (public_ip, new_port), entry saved
In  : dst (public_ip, new_port) -> (private_ip, private_port) from the table
Private : 10/8, 172.16/12, 192.168/16        Max connections per public IP ~ 2^16
No table entry -> inbound packet dropped
```

Practice questions

● **LEVEL 1 · BEGINNER**

Q1. Which fields does NAT change in an outgoing packet?

Q2. Name the three private IPv4 ranges.

Q3. Can an outside host start a connection to a hidden inside host without configuration?

Q4. How does NAT tell apart replies for different inside hosts that share one public IP?

● **LEVEL 2 · EXAM**

Q5. NAT public IP 203.0.113.1. Host 10.0.0.5:4000 connects to 198.51.100.9:80 and NAT assigns port 5001. What does the server see, and what does the reply look like?

Q6. Maximum simultaneous connections through one public IP using NAPT?

Q7. Why must NAT recompute checksums?

Q8. Two inside hosts both use source port 3000. How does NAT cope?

● **LEVEL 3 · GATE**

Q9. A NAT can hold about 65,000 translations. If each inside host keeps 500 connections open, how many hosts can be served?

Q10. Table: (10.0.0.2, 3000) ↔ (203.0.113.1, 6000). A packet arrives for 203.0.113.1:6000 from 198.51.100.7:80. What is delivered, and to whom?

Q11. Why does active FTP have trouble through NAT?

Answers

Q	Solution
1	**Source IP and source port** (and the checksums).
2	**10.0.0.0/8, 172.16.0.0/12, 192.168.0.0/16**.
3	**No**: there is no table entry.
4	By the **external port number** in the table.
5	Server sees **source 203.0.113.1:5001**, destination 198.51.100.9:80. Reply: source 198.51.100.9:80, destination 203.0.113.1:5001, which NAT rewrites to **10.0.0.5:4000**.
6	About $\mathbf{2^{16} = 65{,}536}$ (port field is 16 bits).
7	The IP header checksum covers the source IP, and the TCP/UDP checksum covers a pseudo-header with the IPs and the port fields. Both change.
8	It assigns **different external ports** (for example 6000 and 6001).
9	65,536/500 = 131.07, so at most **131 hosts**.
10	It is delivered to **10.0.0.2:3000** with source still 198.51.100.7:80.
11	The control connection carries an **IP address and port inside the data**; NAT only rewrites headers, so the server is told the useless private address.

Next: Chapter 12 — TCP Segment and Connection Management.

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